KIET Deemed to be University, Ghaziabad

Assignment: UML Activity Diagram (Payment Processing with Swimlanes)

Name: Ashutosh Tiwari | Branch: CSE | Section: B | Semester: V

University Reg. No.: 202401100200098 | University Roll No.: 2400290100098 | Project: Digital Payment Gateway Simulation

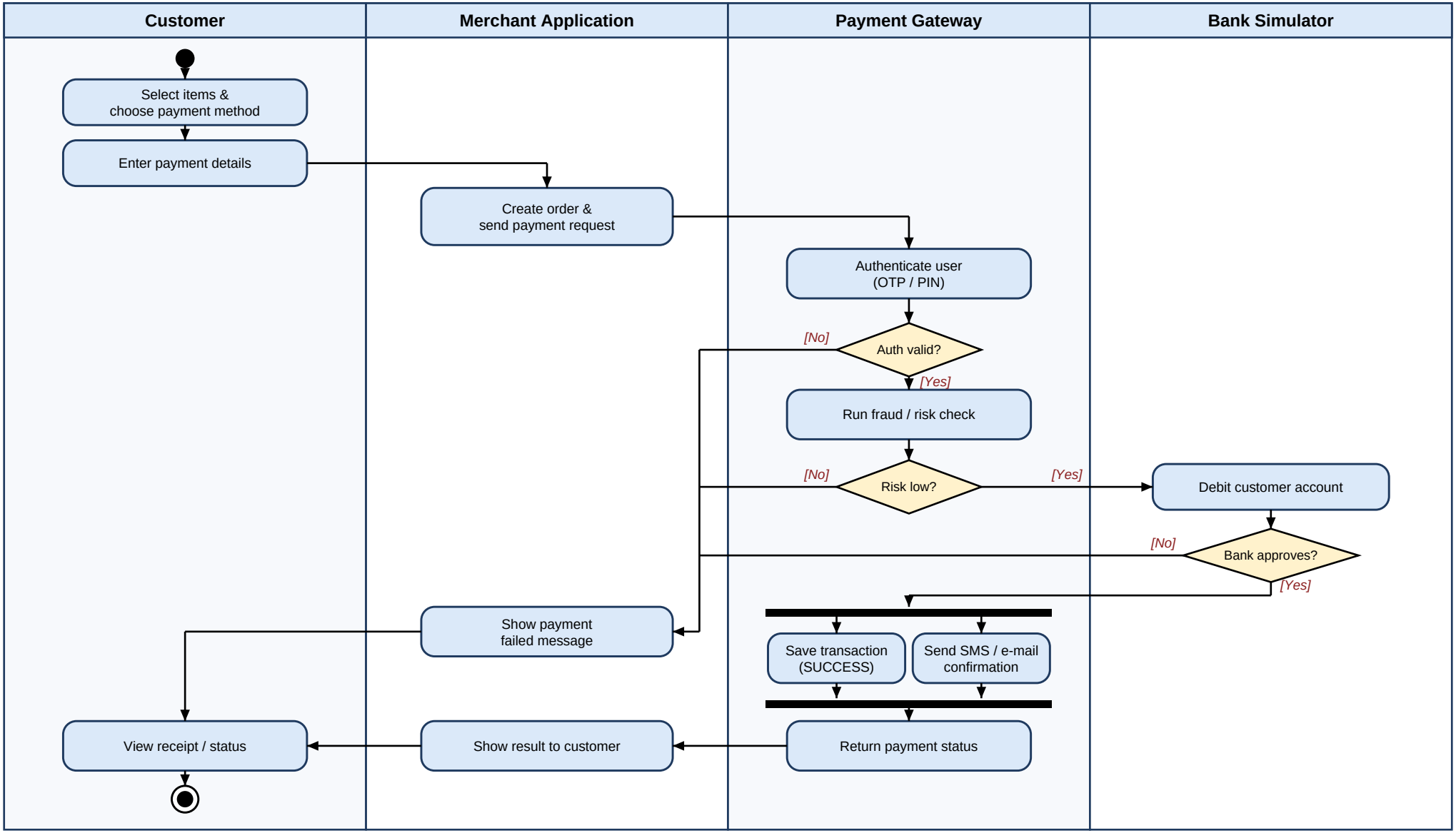

Fig.: Activity Diagram of Payment Processing in the Digital Payment Gateway Simulation

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Assignment: UML Activity Diagram - Activity Description

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Swimlane	Activity / Node	Description
Customer	Select items & choose payment method	Customer picks products and a payment method (Card / UPI / Net Banking / Wallet)
Customer	Enter payment details	Customer types the card number, UPI ID or bank login details
Merchant Application	Create order & send payment request	Merchant creates an order with a unique ID and sends the amount to the gateway
Payment Gateway	Authenticate user (OTP / PIN)	Gateway verifies the user using OTP, PIN or password
Payment Gateway	Decision: Auth valid?	[Yes] continue to risk check; [No] go to the failure path
Payment Gateway	Run fraud / risk check	Checks amount limit, repeated attempts and suspicious behaviour
Payment Gateway	Decision: Risk low?	[Yes] send the request to the bank; [No] decline the payment
Bank Simulator	Debit customer account	Bank simulator checks the balance and tries to debit the amount
Bank Simulator	Decision: Bank approves?	[Yes] payment is approved; [No] payment is declined
Payment Gateway	Fork: Save transaction + Send confirmation	Two parallel activities: store the SUCCESS record and send SMS / e-mail
Payment Gateway	Join, then Return payment status	Both parallel activities finish, then the gateway returns the status to the merchant
Merchant Application	Show result to customer	Merchant shows the success message and receipt (success path)
Merchant Application	Show payment failed message	Merchant shows the failure reason (any [No] path)
Customer	View receipt / status	Customer sees the final result; the activity ends (final node)